NORTHWESTERN POLYTECHNIC
APPLIED RESEARCH & INNOVATION

Faculty Innovation Fund Award
2025/2026

The Faculty Innovation Fund Award supports Northwestern Polytechnic faculty in the early stages of research so that applicants may build their research program to develop scholarly and creative outputs, apply for additional external grants, and/or leverage partnerships in community and industry.

Considerations and Guidelines for Submissions:
Full-time continuing faculty members are eligible to apply as primary or co-investigators.

Budget and budget justifications must be within the award of **up to $5,000** to be considered for review.

Project proposals must:

- clearly outline how funding will be used.
- abide by NWP policies including but not limited to: all Financial policies, Responsible Conduct of Research Policy, Research Involving Human Subjects Policy, Animal Care and Animal Usage Policy, Scholarly Activity Policy, and Conflict of Interest Policy.
- be submitted by Sunday, June 15, 2025 at 11:59 PM.

Award funds:

- will be administered through the Office of Applied Research and Innovation (OARI). OARI must review and pre-approve all expenses to ensure eligibility per the Tri-Agency Guide on Financial Administration and NWP policies.
- must be spent by March 15, 2026 to facilitate institutional fiscal year-end requirements. Expense request forms must be received no later than March 1, 2026.

Please note: funds will not be released for projects involving human participants and/or animals until the primary investigator provides OARI with a letter of approval from the Research Ethics Board and/or Animal Care Committee.

Student Research Assistants:
Hiring Student Research Assistants can enhance your research project while providing valuable experience to students.

Student Research Assistants

- are paid $19/hr plus vacation and stat pay.
- will be hired through the Office of Applied Research and Innovation in collaboration with Human Resources. The recruitment and hiring process for Student Research Assistants can

take several weeks and timing should be considered when building your workplan and timelines.

- will be onboarded by the Office of Applied Research and Innovation. OARI will facilitate submissions of timesheets after hours are verified by the faculty supervisor on the project.
- will work less than 10 hours per week and cannot work when Northwestern Polytechnic is closed (e.g., statutory holidays).
- will be hired on standardized 50- or 100-hour contracts with a contract end date of March 15, 2026. Please ensure renumeration for RA work is included in the overall project budget using the below amounts.

Contract Options:

$1,000 – 50 hours
$2,000 – 100 hours

Please note, Student Research Assistants must:

☒ be enrolled full-time in a degree, diploma, or certificate program at NWP
☐ have completed at least one research methods course
☒ have a competitive GPA

Faculty Innovation Fund Award recipients **are permitted** to apply for up to an additional $2,000 to hire RAs through the Student Research Assistant Program.

Application Assessment:
Applications submitted by Sunday, June 15, 2025 at 11:59 PM will be reviewed by a committee of members of the Applied Research and Innovation Advisory Committee from June 16 to June 29.

Additional information may be requested from the primary investigator as project proposals are evaluated by the committee.

Applications will be assessed on:

- feasibility within given timeframe and budget
- potential for project scaling
- impact on industry/community
- benefit to NWP students

Application Submission:
Please review the Application Checklist below to ensure all required documents are completed. Compile into one PDF and submit to the Office of Applied Research and Innovation at Research@NWPolytech.ca with subject line: Faculty Innovation Fund Award Application.

Application Deadline: **Sunday, June 15, 2025 at 11:59 PM**.

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Post-Award:
Expenses will be processed, based upon pre-approval, through the Office of Applied Research and Innovation. NWP policies and the Tri-Agency Guide on Financial Administration will inform expense approval.

Faculty Innovation Fund Awards cannot be used for capital expenses.

Any equipment purchased with Faculty Innovation Fund Awards remains the property of Northwestern Polytechnic.

Recipients will be expected to submit a funding report detailing findings and next steps for funding applications by Friday, May 15, 2026.

Important Dates:
- May 14, 2025 – application period opens.
- June 15, 2025 – application period closes.
- June 16 to 29, 2025 – application assessments by members of Applied Research and Innovation Advisory Committee.
- June 30 to July 4, 2025 – Faculty Innovation Award Fund recipients notified, and letters of award issued.
- Funding Period from July 15, 2025 to March 15, 2026.
- March 1, 2026 – last day to submit expense requests for approval.
- March 15, 2026 – contracts end for Student Research Assistants.
- May 15, 2026 – Funding and Findings Report submission deadline.

Application Checklist:
Please complete and submit all the below items in a single PDF file to **Research@NWPolytech.ca**.

☒ Application Form
☒ Budget Table
☒ Budget Justification
☒ CVs for all investigators
☒ TCPS-2: CORE 2022 certificates for all investigators

NORTHWESTERN POLYTECHNIC
APPLIED RESEARCH & INNOVATION

Faculty Innovation Fund Award Application Form:

Research Team

Primary Investigator (PI)'s Name: Hanna Yehoshyna
PI Department: Science Department
PI Position: Computer Science Instructor
PI Email: HYehoshyna@nwpolytech.ca
PI Phone: (780) 539-2074

Is this the PI's first time working on a funded research opportunity at NWP: Yes
PI's other active projects: No

Please list all current research team members including co-investigators, students, assistants, faculty supervisors, community organizations, and clients.

Full Name	Email	Project Role	Institutional Affiliation
Mackenzie Beck	5086546@nwpolytech.ca	Software Engineer	Northwestern Polytechnic
-	-	-	-
-	-	-	-

Project Information

Project Title: Leveraging game-based learning to increase students engagement and motivation in Math courses
Summary of Proposal (max 1000 words). Describe the importance of the research topic, the objectives, research methodology, anticipated outcomes, and their expected benefits/impact:

Despite evidence that leveraging gamification in education have the potential to foster student learning through the introduction of goals, competition, and fun learning environments, modern digital educational practices continue to suffer a lack in the level of student motivation and engagement. This has become especially challenging for educational activities related to the practicing of mathematical problems.

Modern students, being representatives of a digital generation, are spoiled with a variety of video game styles on various platforms. Thus, these students are capable to quickly recognize the integration of game elements (like point systems, badges, etc.) with the goal of masking conventional learning activities. As a result, gamification might fail in keeping learners involved and engaged over longer periods of time.

In contrast, game-based learning is grounded in designing learning activities in such a way that video game mechanics (such as game characteristics, patterns of interaction with the game world, game principles that involve rules and procedures) are inherently part of the learning activities themselves.

Essentially, gamification applies game elements, or a game framework, to existing learning activities; while game-based learning designs intrinsically game-like learning activities.

By intelligently designing game mechanics to incorporate mathematical principles and express them in the 'grammar of game design', this study seeks to address the shortcomings of traditional gamification. With this approach students will be encouraged to study mathematics without needing to feel as though they are studying.

The main *objective* of the study is to address the lack of student interest in studying mathematics by enhancing motivation and engagement through the development of effective game-based learning applications.

The *research method* that will be followed is the research and development (R&D) method - a systematic approach used to develop new products or technology. In the context of developing game-based learning applications to increase student learning motivation, an R&D approach can be used to ensure that the application developed is effective, meets educational needs, and is able to increase student learning motivation and engagement in a significant way.

The anticipated outcomes:

The main anticipated output is an educational application prototype, implemented as a 2D movement puzzle game that involves maneuvering a spaceship on the Cartesian plane.

The anticipated outcomes refer to the following:

1) The primary outcome is a tangible game-based app with the aim of enhancing students' interest and engagement in mathematics.

2) Student engagement with the app will result in enhanced skills and proficiency in mathematics seeing that performance in the app is directly tied to the ability of the student to manipulate algebraic expressions.

3) The app has the potential for educational impact by being implemented in educational programs and curriculum. Implementation could lead to improved teaching methods, increased in-class student attention, and problem-solving skills.

4) An eventual outcome would include knowledge transfer products, for example a user manual or teacher guide.

The expected benefits/ impact

The expected benefits and impact of this study include the following:

1) Game-based learning approach to addressing the lack of motivation and engagement of high school students with mathematics
2) By increasing engagement with mathematics, the resultant outcome is improved mathematical skills and grades
3) The development of this application could have an impact on educational curriculum and teaching methods utilized in the mathematics classroom

4) The benefit of utilizing an application of this kind is the engagement with students in a manner relevant to their generation.
5) An estimated benefit is that students will respond more positively to activities related to homework, and have the opportunity to practice Math problems in fun and new ways.

Potential for project scaling.

The project aims to scale from a prototype application with basic functionality to a fully functional educational application.

Scaling would include the following:

- A completed game-based learning application;
- Embedding Roguelike Mechanics (challenging gameplay loop, where players must navigate randomly generated levels, make strategic decisions, and rely on their skill to survive and progress) to ensure replayability and unique experiences with each playthrough;
- Adding Game Assets (animations, textures, sound effects for actions, background music and track variations, etc.);
- Multiplayer.
- Local high school math teachers would be sought out as partners in this project. The product can then be tested with focus groups and further refined based on feedback.
- If there is success with the focus groups, the app can be piloted at a local school to assess efficacy in a learning environment.
- Based on the success of this pilot study, the rollout and adoption of the app in a broader area can be motivated with educational authorities.

With the further integration of advanced technologies such as artificial intelligence and virtual reality, the potential for these applications is even broader, offering a more personal and in-depth experience.

Which of the following does the proposed research involve?

☐ Animals
☐ Human Subjects
☒ None of the above

For projects involving Animals, has an application been made to the Animal Care Committee? No
If no, what is your timeline for submission? NA

For projects involving Human Subjects, has an application been made to NWP's Research Ethics Board? No
If no, what is your timeline for submission? NA

Anticipated project start date: Septermber 15, 2025
Anticipated project end date: March 15, 2026

Please describe your workplan for the duration of the Faculty Innovation Fund Award from July 15, 2025 to March 15, 2026 (max 500 words):

September 2025:
To explore the modern game mechanics and game design strategies to select those elements that help meet the learning objectives carefully
To define the App's core functionality and features.
October 2025
to design the user interface prototype to ensure a game-based App is intuitive, enjoyable, and easy to navigate
to decide on the platform and development tools.
November, December 2025
To design app architecture
and develop a basic version of the App to test and iterate upon.
January 2026, February 2026 (1st part)
To implement a game-based educational application as a 2D movement puzzle game about maneuvering a spaceship on the cartesian plane
February 2026 (2nd part), March 2026
To develop a user guide

What are the expected outcomes or outputs from this project? What are your plans to scale up the project? (max 500 words)

The expected output: A prototype of the game-based learning application
The anticipated outcomes refer to the following:
1) The primary outcome is a tangible game-based app with the aim of enhancing students' interest and engagement in mathematics.
2) Student engagement with the app will result in enhanced skills and proficiency in mathematics seeing that performance in the app is directly tied to the ability of the student to manipulate algebraic expressions.
3) The app has the potential for educational impact by being implemented in educational programs and curriculum. Implementation could lead to improved teaching methods, increased in-class student attention, and problem-solving skills.
4) An eventual outcome would include knowledge transfer products, for example a user manual or teacher guide.

Please describe the funding you plan to apply for to support this project in the future.

Funding Agency	Grant Type	Grant Submission Deadline	Estimated funds requested from funding agency
NA	NA	NA	NA

Budget Table and Justification

Expenses must be considered eligible under the Tri-Agency Guide on Financial Administration.

Please complete and attach "Budget Table – Faculty Innovation Fund Award 2025-26."

Please provide a detailed budget justification that includes all sections included in the Budget Table. Describe in detail all anticipated expenses listed in the Budget Table including descriptions of each line item, how said line items relate to the activities of the project, and verify the cost of line items by describing their calculation or providing quotes/screenshots.

If you have identified costs in Salaries and Benefits, detail all personnel costs including:

- Hourly rates and anticipated hours of work for professional and technical services.
- Note: costs for Student Research Assistants are detailed above.

If you have identified costs in Operating and Equipment, include:

- Detailed cost breakdown by item.
- Note: **Capital purchases are not eligible**.

If you have identified costs in Knowledge Dissemination and Networking,

- Travel costs must comply with NWP's "Travel and Related Expenses Policy." Costs must detail flights/mileage, per diems, and accommodations. Please describe the purpose of a trip and the function of each attendee (ex. field work or presenting at conference).

Note: NWP receives a GST rebate. When calculating GST on goods and services, please use the rebate percentage of 1.67%.

Student Research Assistants

If you have identified that you wish to hire Student Research Assistants, please complete the following section.

What role or type of work do you anticipate for a Student Research Assistant?

☐ Literature Review
☒ Data Collection

☒ Lab Work
☐ Transcription
☐ Drafting Text
☐ Other Knowledge Mobilization
☒ Other: *Please describe.*

Please note, the following are not research activities and may not be assigned to Student Research Assistants:

- Administrative work for courses or course prep
- Organizing course materials
- Personal errands

Are you looking for any specific skills in a Student Research Assistant?

1) Proficiency in C# programming language
2) Knowledge of GDScript scripting language
3) Understanding gameplay mechanics, story development, and level design
4) Understanding of Godot's engine features
5) Confidence in using Git and GitHub
6) Understanding of UI/UX design (Competitor Analysis -SWOT, Empathy Map, User personas, User stories, Scenarios, Storyboards, User Flow)
7) Familiarity with QA fundamentals
8) Familiarity with Godot editor, VS Code IDE
9) Object-oriented Design (including UML) and Development UI/UX design techniques (User Personas, User Stories);
10) Object-oriented Design (UML Use Case Diagram, Class Diagram);
11) QA in Game Development

Please describe the training you will provide to ensure the Student Research Assistant is prepared to assist with the work identified above:

1) UI/UX design techniques (User Personas, User Stories);
2) Object-oriented Design (UML Use Case Diagram, Class Diagram);
3) QA in Game Development

Faculty Innovation Fund Award
2025/2026

Name of Applicant:	Hanna Yehoshyna
Project Title:	d learning to increase students engagement and mo

Proposed Expenditures

Salaries and Benefits	
1) Salaries to non-students:	0
2) Professional and technical services:	$ -
3) Course load reduction for instuctors:	$ -
4) Salaries to students*:	$ 1,000.00
Operating and Equipment	
5) Materials, supplies, and other expeditures:	$ 50.61
6) Equipment:	$ -
Knowledge Dissemination and Networking:	
7) Travel:	$ -
8) Networking meeting:	$ -
9) Workshops and seminars:	$ 504.20
10) Other:	
Total Proposed Expenditures:	$ 1,554.81

*Please see "Student Research Assistants" in application form for totals.

Budget Justification

Name of Applicant: *Hanna Yehoshyna*
Project title: *Leveraging game-based learning to increase students' engagement and motivation in Math courses*

Item Name	***Item Details***	***Description***
	*Personnel costs (**1000 CAD**)*	
Salaries to students ($19/hr) – 50 hours	3 hr	Modern game mechanics and game design strategies exploration
	5 hr	The App's core functionality and features definition (User Personas design, User Stories development, Object-oriented design with UML diagrams)
	3 hr	UI prototype design
	2 hr	Platforms and tools exploration and selection
	5 hr	App architecture design (UML diagrams)
	9 hr	Development of a basic version of the App
	3 hr	QA testing of a basic version of the App
	10 hr	New features implementation and embedding to the prev. step results
	5 hr	QA testing
	5 hr	User guide design
	*Operation and Equipment (**49.78 CAD + 1.67%GST = 50.61 CAD**)*	
Assets for the game-based App	https://azagaya.itch.io/sci-fi-theme $1.99 USD=1.99*1.38=2.75 CAD	A ready-to-use theme for pixel art games made with Godot game engine.
	https://helianthus-games.itch.io/parallax-space-background $6 USD=6*1.38=8.28 CAD	A set of detailed pixel art parallax backgrounds. Also included are 2 star twinkling and 1 comet animations in 2 color schemes to bring additional life to a game's setting
	https://elvgames.itch.io/sci-fi-ui-sounds-fx-pack $2.99 USD=2.99*1.38=4.13 CAD	Sci-Fi UI Sound Effects, Robotics, Consoles, Modern UI

Item Name	***Item Details***	***Description***
	https://untiedgames.itch.io/super-pixel-sci-fi-fx-pack-1 $9.99 USD==9.99*1.38=13.78 CAD	Pixel art effect types, a wide variety of fluid, high-quality animations
	https://guardian5.itch.io/spaceship-megapack $7.55 USD=7.55*1.38=10.42 CAD	Spaceship Megapack has the top-down spaceships to fill game universe (64 unique spacecraft organized within 64x64 pixel tiles on a 512x512 .png image)
	https://free-game-assets.itch.io/resources-for-cyberpunk-topic-pixel-art-3232-icon-pack $7.55 USD=7.55*1.38=10.42 CAD	The necessary resources to create an amazing 2D cyberpunk game project (40 prepared icons on the cyberpunk theme and in pixel art style)
Knowledge dissemination and Networking (***495.92 CAD+ 1.67%GST = 504.2 CAD***)		
Game Development Courses	https://www.udemy.com/course/ultimate-godot-game-bootcamp/?couponCode=ST21MT30625G1 CA $ 49.99 * 2 = 99.98	Ultimate Game Bootcamp- Godot/GDScript/Shader/C# The Hard Way
	https://www.udemy.com/course/godot-ui/?couponCode=ST21MT30625G1 CA $ 61.99 * 2 =123.98	Master Godot UI: Customizing Interfaces with Control Nodes
	https://www.udemy.com/course/godot-vfx-for-games/?couponCode=ST21MT30625G1 CA $ 61.99 * 2 =123.98	Godot - VFX for Games - Beginner to Intermediate
	https://www.udemy.com/course/godot-4-shaders/?couponCode=ST21MT30625G1 CA $ 73.99 * 2 =147.98	Godot 4 Shaders: Craft Stunning Visuals

Hanna Yehoshyna - Curriculum Vitae

Computer Science Instructor

HYehoshyna@nwpolytech.ca

LinkedIn Profile

Orcid Profile

Google Scholar Page

- Enthusiastic, self-motivated, responsible Associate Professor and Researcher with over 20 years of experience teaching and researching Computer Science, Artificial Intelligence, Software Engineering.
- Published research focuses on the Machine Learning, Deep Learning, Knowledge-Based Systems, Computational Linguistics and Recommender Systems.
- Have acted as thesis supervisor for undergraduate and graduate students.
- Experience in preparing and revising courses, designing, developing and evaluating academic programs and curriculums.

WORK EXPERIENCE

Computer Science Instructor 08/2023 -Present (Full-time)

Science Department, School of Applied Science and Technology, Northwestern Polytechnic, AB, Canada

Teach lecture-based courses and conduct laboratory sessions (in a variety of teaching strategies, learning technologies and tools)

- Bachelor's Degree Program "Computer Science":
  - *Software engineering (Requirements analysis, UI/UX design fundamentals, Object-oriented Design with UML, Design Patterns, QA fundamentals)*
  - *Introduction to Artificial Intelligence (Intelligent agents, heuristic and adversarial search, game playing, knowledge representation and reasoning, making decisions, machine learning fundamentals)*
  - *Introduction to File and Database Management (Microsoft Access, PostgreSQL, SQLite, MySQL)*
  - *Algorithms (Graphs and Applications, Analysis of Algorithm: Asymptotic Notations and Basic Efficiency Classes, Algorithm Design Techniques)*
  - *Web Based Internet Technology (HTML, CSS, JavaScript, jQuery, React / Redux fundamentals, PHP and MySQL)*
- Bachelor's Degree Program "Teacher Education North":
  - *Technology Tools for Teaching and Learning*

Curriculum review and development

Prepare and revise courses (including online courses development);

Carry out research activities, preparing and writing scientific papers in field of specialization.

Professor 08/2023 -12/2023 (Part-time)

Faculty of Applied Science and Technology, School of Applied Computing, Sheridan College, ON, Canada

Teaching at the undergraduate level (including online synchronous and asynchronous delivery)

- Honours Baccalaureate Degree Program "Computer Science", "Mobile Computing":
  - *Programming Languages (Principles and Paradigms, C and C++, Object-Oriented Programming Languages: Principles and Paradigms, Functional Programming)*

EDUCATION

Associate Professor
(related to Department of Artificial Intelligence Systems)
State University of Informatics and Artificial Intelligence, Ukraine
2011

Doctor of Philosophy in Artificial Intelligence Systems and Tools
State University of Informatics and Artificial Intelligence, Ukraine
2009

Artificial Intelligence Systems and Tools, Postgraduate Degree
State University of Informatics and Artificial Intelligence, Ukraine
2005 – 2008

Computer systems analyst, Master's Deg.
Donetsk State Institute of Artificial Intelligence, Ukraine
2003 – 2005

Bachelor of Computers Science
Donetsk State Institute of Artificial Intelligence, Ukraine
1999 - 2003

WORK EXPERIENCE

ASSOCIATE PROFESSOR 09/2021 – 08/2023

Department of Information Technology, Odesa Polytechnic National University, Odesa, Ukraine

Teaching at both the graduate and undergraduate levels (including online synchronous and asynchronous delivery)

- Bachelor's Degree Program “Information systems and technology”:
  - *Artificial Intelligence Systems (Data analysis and Machine Learning with Python: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn)*
  - *Deep Learning (TensorFlow, Keras, OpenCV)*
  - *Algorithmization and programming (with C++)*
- Master's Degree Program “Information systems and technology”:
  - *Algorithms and Technologies of Recommendation Systems (with Python)*

Supervise and evaluate project-based courses: *Algorithmization and programming (C++); Algorithms and Technologies of Recommendation Systems (with Python)*

Conduct seminars, conferences and discussion groups with students in field of specialization

Supervise Bachelor's and Master's Degree theses

Participating in the Bachelor's and Master's degree thesis Committee

Prepare and revise courses (including online courses development); design, develop, evaluate programs and curriculums

Organizing and holding an international scientific and practical conferences

Carry out research activities in field of specialization

SOFTWARE DEVELOPMENT TRAINER 05/2018 – 08/2023 (Part-time)

Department of Software Development, IT Academy STEP, https://itstep.org/

Providing instruction on a wide range of Software Development courses and activities: Programming Languages: (*Python, C++, HTML, CSS, JavaScript*), Object-oriented Programming, Object-oriented Design (*UML*), Design Patterns.

Development, supervision and assessing of team projects.

Planning, preparing and evaluating courses.

Code templates designing and development.

Invigilating examinations.

Checking and assessing students' work and giving feedback.

Helping with troubleshooting and resolving issues related to software implementations.

ASSOCIATE PROFESSOR 09/2021 – 08/2023 (Part-time)

Department of Software Development, IT STEP University, Odesa, Ukraine

Teaching at the undergraduate level (Bachelor's Degree Program “Computer science”): Data Mining (*NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn*), Front-end Development (*JavaScript, jQuery, React - Redux fundamentals*)

Advising and mentoring undergraduate students through assignments and projects.

Development of course materials and code templates.

TECHNICAL SKILLS

- Proficiency in *HTML, CSS, JavaScript, C++, Python*
- Familiarity with web development frameworks, libraries (*jQuery, React, Redux fundamentals*)
- Knowledge of Data Wrangling, Data Exploration (*NumPy, Pandas*), Data Visualization Tools (*Matplotlib, Seaborn*)
- Understanding and application of Machine Learning (*Scikit-learn*) and Deep Learning (*TensorFlow, Keras*)
- Solid understanding of UI/UX design (*Competitor Analysis -SWOT, Empathy Map, User personas, User stories, Scenarios, Storyboards, User Flow*)
- Object-oriented Design (including *UML*) and Development
- Knowledge of Data Structures and Algorithms
- Understanding and application of Design Patterns
- Understanding of databases and *SQL*
- Familiarity with QA fundamentals.
- Confidence in using *Git* and *GitHub*
- Familiarity with *Agile, Scrum* and *Jira*
- Proficiency in *Zoom, Microsoft Teams, MS Office Applications* and *CMS WordPress*

LANGUAGES

English

Ukrainian

Russian

WORK EXPERIENCE

ASSOCIATE PROFESSOR 09/2015 - 09/2021 (Full-time)

Department of Information Technology, O.S.Popov Odessa National Academy of Telecommunications, Odesa, Ukraine

Teaching at both the graduate and undergraduate levels:

- Bachelor's Degree Programs "Software engineering", "Computer science":
  - *Data Mining (NumPy, Pandas, Matplotlib, Seaborn)*
  - *Artificial Intelligence Systems (Machine Learning with Python, Scikit-learn)*
  - *Requirements Analysis*
  - *WEB application development (HTML, CSS, JavaScript, jQuery, React, Redux fundamental)*
  - *UI/UX design (Competitor Analysis -SWOT, Empathy Map, User personas, User stories, Scenarios, Storyboards, User Flow)*
- Master's Degree Program "Software engineering":
  - *Algorithms and Technologies of Recommendation Systems (with Python)*

Supervise Bachelor's and Master's Degree theses, participating in the Bachelor's and Master's degree thesis Committee

Prepare and revise course; design, develop, evaluate programs and curriculums.

Participating in Bachelor's Degree Programs "Computer science" and Master's Degree Program "Software engineering" accreditations.

ASSOCIATE PROFESSOR 09/2009 - 09/2015

Department of Artificial Intelligence Systems (09/2009 - 09/2014)

Department of Computer science (09/2014 - 09/2014)

Donetsk National Technical University, Donetsk (Pokrovsk since 2014), Ukraine

Teaching at both the graduate and undergraduate levels:

- Bachelor's Degree Programs "Software engineering", "Computer science":
  - *Data Mining*
  - *Artificial Intelligence Systems*
  - *Web Design*
  - *WEB application development*
- Master's Degree Programs "Software engineering", "Computer science":
  - *Intelligent learning systems and virtual educational environments*
  - *Knowledge engineering and Expert System Development*

QUALIFICATIONS AND ABILITIES

- Teaching experience in a variety of modes (lecture, discussion, active student involvement and laboratory work).
- Implementing modern educational technology and tools in online (including both online synchronous and asynchronous delivery), hybrid and in-person classes
- Tutoring and advising students on courses and academic matters
- Experience in designing courses (for online and in-person classes) with regard for learning outcomes, developing (evaluating) academic programs and curriculums
- Good verbal and written communication skills
- Ability to relate effectively to a wide variety of learners
- Excellent communicator with strong interpersonal, organizational, and problem-solving skills
- Ability to work independently and within a team environment
- Development of research interests with student involvement
- Participating in writing scientific articles, preparing presentations and publications, presenting research results at seminars and scientific conferences
- Passion to learn and improve teaching, technical skills and scientific background.

PANEL ON
RESEARCH ETHICS

Navigating the ethics of human research

TCPS 2: CORE 2022

Certificate of Completion

This document certifies that

Hanna Yehoshyna

successfully completed the Course on Research Ethics based on the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2: CORE 2022)

Certificate # 0001435016

29 May, 2025